

Respiration

THE PROCESS OF RESPIRATION SUPPLIES ENERGY FOR LIFE.

All living things are powered by the energy released by a respiration reaction that takes place inside cells. This reaction needs a supply of oxygen taken from the surrounding air or water.

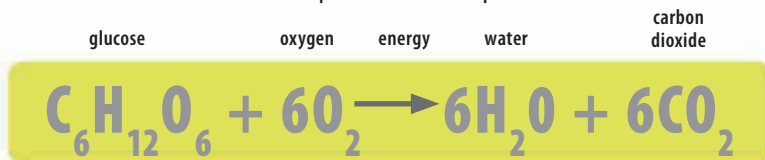
SEE ALSO

Photosynthesis	30–31 >
Combustion	130–131 >
Redox reactions	132–133 >
Energy	170–171 >

Cellular respiration

Every cell produces its own energy by respiration. The process takes place in tiny power plants called mitochondria. A cell that uses a lot of energy, such as a muscle cell, has a large number of these organelles. Respiration is a chemical reaction in which glucose (a sugar and important source of energy) is oxidized (chemically combined with oxygen). As well as energy, the reaction produces carbon dioxide and water.

chemical equation for cellular respiration



▽ Storing and releasing energy

The energy released from respiration is stored by a chemical called adenosine triphosphate (ATP). The energy is used to add a phosphate (P) to adenosine diphosphate (ADP), to store energy. When needed elsewhere in the cell, the phosphate breaks off and releases the energy.



energy
gained



energy
released

▽ Anerobic respiration

If the cell cannot get enough oxygen to power respiration, it does it anerobically, meaning “without air.” This process produces lactic acid as a result, which is what makes hard-working muscles burn with fatigue. Anerobic respiration releases only part of the energy in glucose, but the rest is released when oxygen is available again.

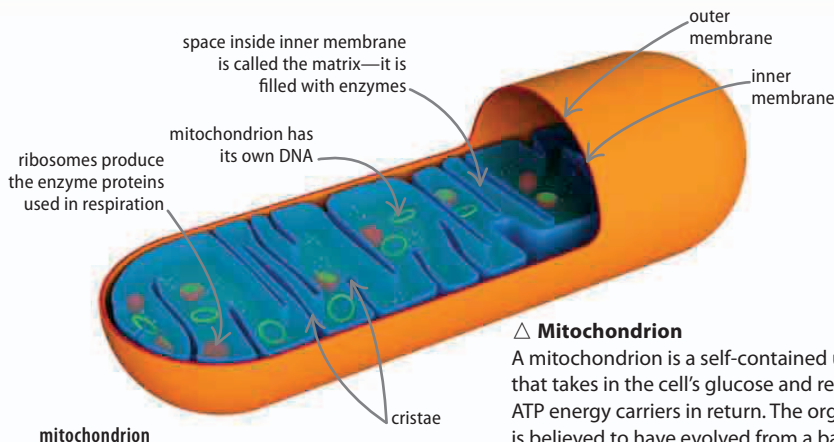
glucose

lactic acid



Mitochondrion

A mitochondrion is surrounded by an outer membrane, similar to the one around a cell. There is another membrane inside that is folded in on itself. The folded areas are called cristae. The main enzymes that control the production of ATP are bonded to the inner membrane. This is where respiration happens. The cristae increase the surface area of the inner membrane, maximizing the space for the enzymes.



△ Mitochondrion

A mitochondrion is a self-contained unit that takes in the cell's glucose and releases ATP energy carriers in return. The organelle is believed to have evolved from a bacterium that began to live inside larger cells.